Ramadan-Aware Short-Term Electricity Load Forecasting:

Benchmarking Time-Series Foundation Models Under Hijri-Calendar Regime Shifts

Capstone Project Proposal

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1 Introduction and Motivation

Short-term load forecasting (STLF)—predicting electrical demand 24 hours ahead—underpins unit commitment, reserve scheduling, and renewable integration. In fuel-importing MENA economies, forecast error translates directly into imported-fuel costs and load-shedding risk. Yet dominant STLF benchmarks (GEFCom, UCI, ENTSO-E) use Gregorian-calendar, temperate-climate data and encode none of the regime-altering dynamics present in Muslim-majority grids: **(i) Ramadan regime shifts**—abrupt restructuring at Hijri month boundaries with iftar spikes, suhoor pre-dawn peaks, compressed commercial hours, and collapsed weekday/weekend patterns; and **(ii) extreme-summer heatwave demand**—sustained T > 40°C driving AC loads >60% of peak with highly nonlinear load–temperature behaviour above ~35°C.

Time-series foundation models (TSFMs)—Chronos, TimesFM, Moirai, Time-MoE—have emerged as competitive STLF candidates (Meyer et al., 2024), yet **no published work evaluates TSFMs under Hijri-calendar structural breaks**. A cautionary result in electricity-*price* forecasting found a simple MSTL baseline undefeated by any TSFM (Olivares et al., 2024), suggesting pretrained representations may fail where regime mismatch is greatest. We propose the first systematic zero-shot TSFM benchmark on a Ramadan-exhibiting grid, with controlled ablations isolating Hijri-feature value.

2 Problem Formulation

Let y_t = aggregate national load (MW) at hour t. We forecast:

$$y\blacksquare_{t+24} = f(y_{t-L+1:t} , x_{t+24})$$

where L is context length and x_t comprises three covariate groups:

Group	Features	Encoding
Meteorological	2m temp, dewpoint, wind, SSRD	ERA5 hourly; pop-weighted avg
Gregorian	Hour, day-of-week, month, holidays	Cyclical sin/cos; binary
Hijri	Ramadan ind., days-into-Ramadan, Eid	Binary + ordinal; Umm al-Qura

Table 1. Covariate groups. The Ramadan indicator is the central experimental variable.

3 Data

Load. Hourly national load from EPIAS (Energy Exchange Istanbul), Jan 2018 – Mar 2025 (~63k observations). Turkey: public data, Ramadan effects, macro-scale (~90 GW peak). **Weather.** ERA5 at 0.25° (Copernicus CDS): 2m temp, dewpoint, 10m wind, SSRD; population-weighted over Istanbul, Ankara, Izmir, Bursa, Antalya. **Splits.** Train: 2018–2022; Validation: 2023; Test: 2024-01 – 2025-03 (covers Ramadan 2024 and 2025). No temporal leakage.

4 Methods

All models forecast $y\blacksquare_{t+24}$ with identical covariate access; the only controlled variable across ablations is Hijri-feature inclusion.

4.1 Classical Baselines

MSTL + ETS. Decomposition into trend + seasonal components (periods 24, 168, 8766) via LOESS; residual forecast with ETS(A,N,N). During Ramadan windows the daily seasonal component is re-estimated from Ramadan-only history (explicit regime switch). **SARIMAX.** $SARIMA(p,d,q)(P,D,Q)_{24}$ with exogenous regressors; order selection via AICc stepwise search.

4.2 ML and Deep-Learning Baselines

LightGBM. Features: 168 lagged loads, all covariates, 7/30-day rolling mean and std. Optuna-tuned (50-trial TPE). **PatchTST.** Channel-independent Transformer (patch P=16, stride S=8, 3 layers, 4 heads, d=128). MSE loss, AdamW (lr=1e-4), cosine schedule, early stopping.

4.3 Time-Series Foundation Models (Zero-Shot Only)

Model	Architecture	Covariate Handling
Chronos-Bolt (Large)	T5 encoder-decoder; tokenized TS	None (univariate)
TimesFM 2.0	Decoder-only patched Transformer	Static + dynamic covariates
Moirai-1.1 (Large)	Masked encoder (any-variate)	Multi-variate channels
Time-MoE (Large)	Mixture-of-Experts decoder	Channel-independent

Table 2. TSFMs — **zero-shot only, no fine-tuning**. For covariate-capable models (TimesFM, Moirai), Hijri features injected as dynamic covariates. For univariate models, Ramadan ablation compares raw output vs. post-hoc LightGBM residual correction on Ramadan errors.

5 Evaluation Protocol

Metrics. MAE (MW), MAPE (%), RMSE (MW), MASE (vs. seasonal-naive lag-168). **Statistical testing.** Diebold-Mariano test with Newey-West HAC standard errors; alpha=0.05, Holm-Bonferroni corrected. **Regime-conditional analysis.** All metrics reported in aggregate and stratified across four operating regimes:

Regime	Definition	Diagnostic Goal
Normal	Non-Ramadan, T_max < 35°C	Baseline accuracy
Ramadan	Within Ramadan month	Hijri-feature marginal value

Regime	Definition	Diagnostic Goal
Heatwave	Non-Ramadan, T_max >= 35°C, >= 3 consecutive days	Nonlinear weather-load stress
Compound	Ramadan AND T_max >= 35°C	Maximum forecast difficulty

Table 3. Operating-regime stratification for conditional error analysis.

6 Ablation Design

A. Ramadan-indicator inclusion (primary). Every model evaluated ± Hijri features; per-regime metric delta isolates explicit Ramadan-awareness value. For univariate TSFMs: raw vs. post-hoc residual correction. **B. Heatwave interaction.** Ramadan × T_{max} interaction feature tests compound-regime encoding benefit. **C. Context-length sensitivity.** TSFMs evaluated at $L \in \{96, 168, 336, 720\}$ hours to test whether longer contexts spanning Ramadan transitions improve regime-shift capture.

7 Timeline (4 Weeks)

Wk	Activities
1	EPIAS + ERA5 acquisition, preprocessing, Hijri feature engineering, exploratory data analysis
2	Implement & evaluate MSTL, SARIMAX, LightGBM, PatchTST on train/val/test splits
3	Zero-shot TSFM evaluation (4 models); execute ablations A, B, C across all methods
4	DM tests, regime-conditional tables, error decomposition analysis, final report & code release

Table 4. Four-week project schedule.

8 Expected Contributions

C1. First published zero-shot TSFM benchmark under Hijri-calendar regime shifts. **C2.** Open-source pipeline (EPIAS + ERA5 + Hijri features + evaluation harness) as a reusable MENA-grid STLF benchmark. **C3.** Regime-conditional error decomposition identifying when classical models dominate TSFMs and vice versa. **C4.** Practical post-hoc Ramadan-correction recipe for off-the-shelf foundation models.

References

Ansari, A. F., et al. (2024). Chronos: Learning the Language of Time Series. *arXiv:2403.07815*.

Bandara, K., Hyndman, R. J., & Bergmeir, C. (2021). MSTL: Seasonal-Trend Decomposition for Multiple Seasonal Patterns. *Int. J. Op. Res.*

Das, A., et al. (2024). A decoder-only foundation model for time-series forecasting. *ICML 2024*.

Diebold, F. X. & Mariano, R. S. (2002). Comparing Predictive Accuracy. *JBES*, 20(1), 134–144.

Meyer, M., et al. (2024). Benchmarking TSFMs for Short-Term Household Load Forecasting. *arXiv:2410.09487*.

Nie, Y., et al. (2023). A Time Series is Worth 64 Words: Long-term Forecasting with Transformers. *ICLR 2023*.

Olivares, K. G., et al. (2024). Are TSFMs Ready for Electricity Price Forecasting? *arXiv preprint*.

Woo, G., et al. (2024). Moirai: A Time Series Foundation Model for Universal Forecasting. *ICML 2024*.